

COMPREHENSIVE PRACTICE WORKSHEET

Class IX — Mathematics | Chapter 1: Number System

Time Allowed: 3 Hours

Maximum Marks: 100

Student Name: _____

General Instructions:

- This worksheet consists of two main sections: **Section A** (50 MCQs, 1 mark each) and **Section B** (20 Subjective Questions, 50 marks total).
- All questions are compulsory.
- Use of calculators is strictly prohibited.

SECTION A: Multiple Choice Questions (50 Questions × 1 Mark = 50 Marks)

1. Every rational number is a:

- | | |
|--------------------|------------------|
| (a) Natural number | (b) Integer |
| (c) Real number | (d) Whole number |

2. Between any two rational numbers, there exist:

- | | |
|--------------------------------------|---------------------------------|
| (a) No rational number | (b) Exactly one rational number |
| (c) Infinitely many rational numbers | (d) Only integer numbers |

3. The decimal representation of a rational number cannot be:

- | | |
|-----------------------------------|-------------------------------|
| (a) Terminating | (b) Non-terminating repeating |
| (c) Non-terminating non-repeating | (d) Terminating recurring |

4. Which of the following is an irrational number?

- | | |
|-----------------|---------------------|
| (a) $\sqrt{16}$ | (b) $\sqrt{(12/3)}$ |
| (c) $\sqrt{7}$ | (d) $\sqrt{81}$ |

5. The value of $0.333\dots$ in the form of p/q , where p and q are integers and $q \neq 0$, is:

- | | |
|------------|-------------|
| (a) $1/3$ | (b) $3/10$ |
| (c) $1/33$ | (d) $3/100$ |

6. The sum of a rational and an irrational number is always:

- | | |
|-----------------------|--------------------------|
| (a) A rational number | (b) An irrational number |
| (c) An integer | (d) A whole number |

7. Which of the following numbers is represented by the decimal $0.1010010001\dots$?

- | | |
|-----------------------|---------------------|
| (a) Natural number | (b) Rational number |
| (c) Irrational number | (d) Integer |

8. The value of $(256)^{0.16} \times (256)^{0.09}$ is:

- | | |
|--------|------------|
| (a) 4 | (b) 16 |
| (c) 64 | (d) 256.25 |

9. Simplify: $\sqrt{(3^{-2})}$:

- | | |
|-----------|-----------|
| (a) $1/3$ | (b) 3 |
| (c) -3 | (d) $1/9$ |

10. The rationalizing factor of $1 / (\sqrt{7} - 2)$ is:

- (a) $\sqrt{7} - 2$
- (b) $\sqrt{7} + 2$
- (c) $\sqrt{7}$
- (d) $2 - \sqrt{7}$

11. If $x = 2 + \sqrt{3}$, then the value of $x + 1/x$ is:

- (a) 2
- (b) 4
- (c) $2\sqrt{3}$
- (d) 0

12. Which of the following is equal to x ?

- (a) $x^{12/7} - x^{5/7}$
- (b) $\sqrt{(x^{12})^{1/6}}$
- (c) $(x^3)^{2/3}$
- (d) $x^{12/7} \times x^{7/12}$

13. The value of $\sqrt[3]{2}$ is equal to:

- (a) $2^{1/6}$
- (b) 2^6
- (c) $2^{1/5}$
- (d) 2^5

14. The product of $\sqrt[3]{2} \times \sqrt[4]{2} \times \sqrt[12]{32}$ is:

- (a) $\sqrt{2}$
- (b) 2
- (c) 4
- (d) 12

15. Express 0.4777... (0.47) in the form p/q :

- (a) $47/99$
- (b) $43/90$
- (c) $47/90$
- (d) $9/43$

16. The value of $\sqrt{(10)} \times \sqrt{(15)}$ is equal to:

- (a) $6\sqrt{5}$
- (b) $5\sqrt{6}$
- (c) 25
- (d) $10\sqrt{5}$

17. Which of the following is correct?

- (a) $\sqrt{2} + \sqrt{3} = \sqrt{5}$
- (b) $(\sqrt{2})(\sqrt{3}) = \sqrt{6}$
- (c) $\sqrt{2} + 2 = 2\sqrt{2}$
- (d) None of these

18. The number π (pi) is:

- (a) A rational number
- (b) An irrational number
- (c) An integer
- (d) A prime number

19. The value of $22/7$ is:

- (a) An irrational number
- (b) A rational number
- (c) Neither rational nor irrational
- (d) An integer

20. Simplify: $(\sqrt{5} + \sqrt{2})^2$

- (a) $7 + 2\sqrt{10}$
- (b) $7 + \sqrt{10}$
- (c) $10 + 2\sqrt{7}$
- (d) 7

21. The value of $(64)^{-1/3}$ is:

- (a) $1/4$
- (b) 4
- (c) -4
- (d) $1/16$

22. Find a rational number between $1/4$ and $1/3$:

- (a) $7/24$
- (b) $5/12$
- (c) $1/2$
- (d) $2/5$

23. $(3 + \sqrt{3})(3 - \sqrt{3})$ is equal to:

- (a) 0
- (b) 6
- (c) 9
- (d) 3

24. The simplest form of $0.001001001\ldots$ ($0.00\overline{1}$) in p/q form is:

- (a) $1/999$
- (b) $1/99$
- (c) $1/1000$
- (d) $1/9$

25. Which of the following is true?

- (a) Sum of two irrationals is always irrational
- (b) Product of two irrationals is always irrational
- (c) Sum of two irrationals can be rational or irrational
- (d) Square of an irrational number is always rational

26. Rationalize the denominator of $1/\sqrt{5}$:

- (a) $\sqrt{5}/5$
- (b) $5/\sqrt{5}$
- (c) $1/5$
- (d) $\sqrt{5}$

27. The value of $(81)^{3/4}$ is:

- (a) 9
- (b) 27
- (c) 3
- (d) 81

28. Which of the following numbers is an integer?

- (a) $\sqrt{2}$
- (b) $\sqrt{4}$
- (c) $\sqrt{3}$
- (d) $\sqrt{5}$

29. The product of $\sqrt{3}$ and $\sqrt{27}$ is:

- (a) 9
- (b) $3\sqrt{3}$
- (c) 81
- (d) 3

30. If $2^x = 32$, then x is equal to:

- (a) 3
- (b) 4
- (c) 5
- (d) 6

31. The set of natural numbers is represented by:

- (a) W
- (b) Z
- (c) N
- (d) Q

32. The smallest natural number is:

- (a) 0
- (b) 1
- (c) -1
- (d) Infinite

33. The exponent form of $\sqrt[4]{x^3}$ is:

- (a) $x^{4/3}$
- (b) $x^{3/4}$
- (c) $x^{1/12}$
- (d) x^{12}

34. Simplify: $(2^{2/3}) \times (2^{1/3})$

- (a) 2
- (b) $2^{2/9}$
- (c) 1
- (d) 4

35. What is the conjugate of $3 - \sqrt{5}$?

- (a) $-3 - \sqrt{5}$
- (b) $3 + \sqrt{5}$
- (c) $\sqrt{5} - 3$
- (d) $-3 + \sqrt{5}$

36. Decimal expansion of $1/7$ is:

- (a) Terminating
- (b) Non-terminating repeating
- (c) Non-terminating non-repeating
- (d) None of these

37. Value of $(1^3 + 2^3 + 3^3)^{1/2}$ is:

- (a) 6
- (b) 36
- (c) 12
- (d) 18

38. An example of a non-terminating non-recurring decimal is:

- (a) 0.123123...
- (b) 0.101101110...
- (c) 0.45
- (d) 0.333...

39. Value of $(\sqrt{8} - \sqrt{2})$ is:

- (a) $\sqrt{6}$
- (b) $\sqrt{2}$
- (c) 2
- (d) 0

40. The order of the surd $\sqrt[4]{7}$ is:

- (a) 7
- (b) 4
- (c) $1/4$
- (d) 1

41. Express 0.99999... in standard p/q form:

- (a) $99/100$
- (b) $9/10$
- (c) 1
- (d) $9/99$

42. If $x^2 = 5$, then x is:

- (a) Rational
- (b) Irrational
- (c) Integer
- (d) Complex

43. Which of the following is equal to 1?

- (a) 0^1
- (b) 5^0
- (c) $1^0 / 0^1$
- (d) Both (b) and (c)

44. Rationalize: $1 / (\sqrt{3} + \sqrt{2})$

- (a) $\sqrt{3} - \sqrt{2}$
- (b) $\sqrt{3} + \sqrt{2}$
- (c) $1 / (3 - 2)$
- (d) $\sqrt{6}$

45. If $a = 2$ and $b = 3$, then $(a^b + b^a)^{-1}$ is:

- (a) 17
- (b) $1/17$
- (c) $1/71$
- (d) -17

46. Division of two irrational numbers is:

- (a) Always rational
- (b) Always irrational
- (c) Either rational or irrational
- (d) Always an integer

47. $\sqrt[3]{125}$ is equal to:

- (a) 5
- (b) 25
- (c) $5\sqrt{5}$
- (d) $1/5$

48. Multiplicative inverse of $\sqrt{5}$ is:

- (a) $-\sqrt{5}$
- (b) $1/\sqrt{5}$
- (c) 5
- (d) $\sqrt{5}/5$

49. Which of the following is a pure surd?

- (a) $2\sqrt{3}$
- (b) $\sqrt{12}$
- (c) $5\sqrt{2}$
- (d) $3\sqrt{7}$

50. If $(3/5)^x (5/3)^{2x} = 125/27$, then x is:

- (a) 1
- (b) 2
- (c) 3
- (d) 4

SECTION B: Subjective Questions (20 Questions = 50 Marks)

1. Find five rational numbers between $3/5$ and $4/5$.

[2 Marks]

2. Express $1.272727\ldots$ ($1.\overline{27}$) in the form p/q , where p and q are integers and $q \neq 0$.

[2 Marks]

3. Show how $\sqrt{5}$ can be represented on the real number line using geometrical construction steps.

[2 Marks]

4. Rationalize the denominator of $1 / (7 + 3\sqrt{2})$.

[2 Marks]

5. Simplify the expression: $(\sqrt{3} + \sqrt{7})^2 + (\sqrt{3} - \sqrt{7})^2$.

[2 Marks]

6. Find the value of a and b if $(3 + \sqrt{7}) / (3 - \sqrt{7}) = a + b\sqrt{7}$.

[2 Marks]

7. Simplify: $(81/16)^{-3/4} \times [(25/9)^{-3/2} \div (5/2)^{-3}]$.

[2 Marks]

8. If $x = 3 + 2\sqrt{2}$, find the value of $x^2 + 1/x^2$.

[2 Marks]

9. Express $0.2353535\ldots$ ($0.\overline{235}$) in the form p/q .

[2 Marks]

10. Geometrically represent $\sqrt[3]{9.3}$ on the number line. Write down the step-by-step procedure.

[3 Marks]

11. Evaluate: $[(64/125)^{-2/3} + (256/625)^{-1/4} + (3/7)^0]$.

[3 Marks]

12. If $x = (\sqrt{5} + \sqrt{2}) / (\sqrt{5} - \sqrt{2})$ and $y = (\sqrt{5} - \sqrt{2}) / (\sqrt{5} + \sqrt{2})$, find the value of $x^2 + y^2 + xy$.

[3 Marks]

13. Prove that $1 / (1 + x^{b-a} + x^{c-a}) + 1 / (1 + x^{a-b} + x^{c-b}) + 1 / (1 + x^{a-c} + x^{b-c}) = 1$.

[3 Marks]

14. Simplify by rationalizing the denominators: $1 / (1 + \sqrt{2}) + 1 / (\sqrt{2} + \sqrt{3}) + 1 / (\sqrt{3} + \sqrt{4})$.

[3 Marks]

15. If $2^a = 3^b = 6^c$, prove that $c = ab / (a + b)$ or $1/c = 1/a + 1/b$.

[3 Marks]

16. State whether the following statements are true or false with proper mathematical justification: [3 Marks]

(i) Every irrational number is a real number.

(ii) Every point on the number line is of the form $\sqrt{m}$, where m is a natural number.

17. Find the values of a and b if $(7 + 3\sqrt{5})/(3 + \sqrt{5}) - (7 - 3\sqrt{5})/(3 - \sqrt{5}) = a + b\sqrt{5}$. [3 Marks]

18. Arrange the following surds in ascending order of magnitude: $\sqrt[3]{2}$, $\sqrt[4]{3}$, $\sqrt[6]{5}$. [3 Marks]

19. If $x = 1/(2 - \sqrt{3})$, find the value of $x^3 - 2x^2 - 7x + 5$. [3 Marks]

20. Solve for x : $4^{x-1} \times 0.5^{3-2x} = (1/8)^{-x}$. [3 Marks]